

Spark Engine Technical Report: LCF Anomaly Investigation

Luminous Dynamics

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Version: 0.1.0 **Original date:** 2026-02-05 **Revised:** 2026-07-07 **Status:** Physics-Honest Assessment **Package:** spark-engine on crates.io · github.com/Luminous-Dynamics/spark-engine

Revision note (2026-07-07): this report was updated to reflect the current code-base (99 tests, up from 42) and two capabilities added since the original draft — a Bayesian belief model over the five candidate hypotheses (`bayesian.rs`) and an expected-information-gain experiment ranker (`optimal_experiment.rs`). Section 5's static priority table is now superseded by a regenerable, decision-theoretic ranking — see §5.1. One numeric table from the original draft (§2.3) is flagged as illustrative rather than verified; see the caveat there. No conclusion about the physics itself has changed.

Executive Summary

This report presents a rigorous, physics-honest analysis of Lattice Confinement Fusion (LCF) using the Spark Engine simulation framework. Our key findings:

1. **Standard Gamow physics predicts $Q \ll 1$** at room temperature. LCF cannot produce net energy under known physics.
2. **A 40-order-of-magnitude gap exists** between NASA's observed neutron rate ($\sim 10^3$ n/s) and Gamow predictions ($\sim 10^{-38}$ n/s).
3. **The most plausible explanation** is transient hot spots from X-ray absorption, which could locally achieve temperatures of $\sim 3,500\text{K}$.
4. **LCF may still be valuable** as a compact neutron source, even without energy gain.

5. (New, 2026-07-07) The single best next experiment, by expected information gain per dollar, is the Hydrogen Control (~\$10K, 1 month) — not the Temperature Mapping study this report originally emphasized. See §5.1.
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1. Standard Physics Baseline

1.1 Gamow Integration

The D-D fusion rate is calculated using the full Gamow integral:

$$\langle \sigma v \rangle = \sqrt{8/\pi\mu} \cdot (kT)^{-3/2} \cdot \int \sigma(E) \cdot E \cdot \exp(-E/kT - E_G/\sqrt{E} + U_e/kT) dE$$

Where: - μ = reduced mass (≈ 1 amu for D-D) - E_G = Gamow energy = $31.38 \text{ keV}^{0.5}$ for D-D - U_e = electron screening energy

1.2 Room Temperature Results

Parameter	Value	Unit
Temperature	300	K
Screening (Pd)	309	eV
	5.3×10^{-90}	cm^3/s
Gamow peak	5.0	eV
Q factor	1.8×10^{-57}	-

Conclusion: At room temperature, is negligible. Standard physics predicts essentially zero fusion.

1.3 Temperature Dependence

Temperature (K)	(cm^3/s)	Q Factor
300	5.3×10^{-90}	10^{-57}
1,000	2.9×10^{-64}	10^{-31}
10,000	2.6×10^{-29}	10^{+4}
100,000	1.2×10^{-16}	10^{17}

The rate increases by ~25 orders of magnitude per factor of 10 in temperature. $Q > 1$ requires $T > 10^7 \text{ K}$ under standard physics.

2. The NASA Anomaly

2.1 Observation Summary

NASA Glenn Research Center (Steinetz et al., 2020, Phys. Rev. C) reported:

- **Neutron rate:** $\sim 10^3$ n/s
- **Neutron energy:** 2.45 MeV (consistent with D-D)
- **Host material:** ErD_2 on TiD_2
- **Trigger:** 12 keV bremsstrahlung X-rays
- **Controls:** H control showed no signal

2.2 Gap Analysis

Metric	Value
Observed rate	10^3 n/s
Gamow predicted	4.3×10^{-38} n/s
Gap factor	2.3×10^{40}
Gap orders	40
Classification	Extraordinary

2.3 What Could Explain the Gap?

Enhancement Source	Factor	Cumulative
Electron screening (309 eV)	10^{200}	10^{200}
Phonon enhancement (2 modes)	10^2	10^{202}
Hot spot ($10\times$ T)	10^{10}	10^{212}
Unexplained	10^{-172}	10^{40}

The combined known enhancements far exceed the gap, but each is applied at its theoretical maximum. In practice, these enhancements may not combine multiplicatively.

Honesty caveat (added 2026-07-07): this table's numbers could not be reproduced from the current `screening_enhancement()` implementation (`physics.rs`) at any physically relevant energy for this system — the function returns the trivial value 1.0 below 10 eV collision energy by design, and is capped at 10^6 above that (added later, per `clippy::manual_clamp` cleanup 2026-07-07). The 10^{200} screening figure appears to be from an earlier, uncapped, illustrative back-of-envelope exercise evaluated outside the function's intended domain, not a live-computed result. Treat this table as **narrative intuition about orders of magnitude, not a verified calculation** — which is consistent with the original prose above already flagging the multiplicative-combination assumption as questionable. Recomputing it rigorously (correct energy convention, no artificial floor/ceiling) is open work, not done here.

3. Hypothesis Evaluation

3.1 Hot Spots (Most Plausible)

Mechanism: X-ray absorption deposits energy in ~ 10 nm radius, creating transient regions with $T \gg T_{\text{bulk}}$ for ~ 0.1 ps.

Predictions: - Rate scales with X-ray intensity - Weak bulk temperature dependence
- Burst structure following trigger pulses

Required conditions: - Peak temperature: $\sim 3,500$ K ($12\times$ room temp) - Cooling time: ~ 0.1 ps - Plausibility: **HIGH** - thermally achievable

3.2 Phonon Cascade (Possible)

Mechanism: Multiple optical phonons coherently add their energy (~ 56 meV each) to the D-D center-of-mass.

Predictions: - Rate increases with acoustic drive - Resonances at phonon frequencies
- Strong temperature dependence

Required conditions: - Coherent phonons needed: >50 - Plausibility: **LOW** - maintaining coherence of 50+ modes is difficult

3.3 Super-Screening (Unlikely)

Mechanism: Electron screening energy exceeds measured values due to collective effects at high loading.

Predictions: - Rate correlates with D/M loading - Should be observable in accelerator experiments

Required conditions: - Screening needed: $\sim 100,000$ eV (vs. measured 309 eV) - Plausibility: **VERY LOW** - $300\times$ higher than any measurement

3.4 Measurement Error (Cannot Rule Out)

Mechanism: Signal is not from D-D fusion but from background, cosmic rays, or electronic noise.

Predictions: - Signal persists with H instead of D - No correlation with trigger

Required conditions: - NASA's H control passed - Plausibility: **LOW** but independent replication needed

4. Literature Survey

4.1 Database Summary

Statistic	Value
Total observations	11
Average credibility	6.3/10
High-credibility (≥ 7)	4
With neutron detection	2
With excess heat claims	5

4.2 Most Credible Observations

1. **Raiola 2004** (9.0/10) - Screening measurement, confirmed by multiple labs
2. **Steinetz 2020** (8.5/10) - NASA neutron observation with controls
3. **Czerski 2004** (8.5/10) - Screening in Ti, confirmed
4. **Huizenga 1993** (7.0/10) - Skeptical review, documents failures

4.3 Screening Energy Measurements

Material	U _e (eV)	Reference
Pd	309 ± 12	Raiola 2004
Pt	320 ± 15	Raiola 2004
Ti	363 ± 25	Czerski 2004
Er	~800	Steinetz 2020 (estimated)

These are anomalously large compared to the Debye model prediction (~25 eV) but are well-established by accelerator measurements.

5. Recommended Experimental Program

5.1 Superseding this section: the live, decision-theoretic ranking (added 2026-07-07)

The phase/priority tables below (§5.1–5.3 as originally written) reflect **hand-set priorities** from the initial design. Since then, two modules make this decision-theoretic and regenerable rather than fixed:

- `bayesian.rs` — a probability distribution (`HypothesisBelief`) over the five candidate explanations (hot spots, phonon cascade, super-screening, lattice-modified tunneling, measurement error), with a deliberately skeptical default prior (measurement error weighted highest, 0.40).

- `optimal_experiment.rs` — ranks every candidate experiment by **expected information gain (EIG) per dollar** against that belief, and derives a budget-constrained sequence.

Regenerate the live ranking with:

```
cargo run --example optimal_experiment_proposal 500000 # budget in USD
```

As of this writing (see `docs/DISCRIMINATING_EXPERIMENT_PROPOSAL.md`), the ranking **differs from the static table below in its top pick**: the **Hydrogen Control** experiment — \$10K, 1 month — ranks #1 at 9.71 bits of information gain per \$100K, ahead of the NASA Replication and Temperature Mapping experiments the original static priorities emphasized. This makes physical sense in hindsight: it is the cheapest experiment that cleanly separates “real fusion” from “measurement artifact,” the single largest source of belief uncertainty under the skeptical prior. The static tables below are kept for historical/narrative context but the generated proposal is now the source of truth for sequencing.

Phase 1: Validation (\$200K, 6 months)

Goal: Confirm the anomaly is real and from D-D fusion.

Experiment	Cost	Priority
NASA Replication	\$75K	100%
H Control	\$10K	100%
Neutron Spectroscopy	\$50K	95%

Go/No-Go: All three criteria must pass: - Observe >100 n/s with D - H control shows <1 n/s - Neutron energy = 2.45 ± 0.3 MeV

Phase 2: Mechanism (\$300K, 12 months)

Goal: Discriminate between hypotheses.

Experiment	Cost	Priority
Temperature Mapping	\$80K	90%
Acoustic Drive	\$40K	80%
Host Material Survey	\$60K	85%

Go/No-Go: Identify plausible mechanism.

Phase 3: Optimization (\$200K, 12 months)

Goal: Maximize neutron output for applications.

Experiment	Cost	Priority
Scale-Up	\$50K	70%
Lifetime	\$30K	65%
Efficiency	\$40K	60%

Go/No-Go: Achieve $>10^6$ n/s sustained.

6. Path Forward: Compact Neutron Source

Even with $Q \ll 1$, LCF may be valuable as a neutron source.

6.1 Application Requirements

Application	Rate (n/s)	LCF Viable?
Research	10^4	Yes (if anomaly real)
NAA	10^6	Possible with scale-up
BNCT	10^9	Challenging
Security	10^8	Challenging

6.2 Competitive Comparison

Source	Rate (n/s)	Cost	Portability
Cf-252	$2.3 \times 10^6/\mu\text{g}$	\$50K/ μg	Portable
D-D accelerator	10^8	\$200K	Transportable
D-T accelerator	10^{10}	\$300K	Transportable
LCF (if anomaly)	$10^5/\text{cm}^3$	$\sim \$20\text{K}/\text{cm}^3$	Portable

LCF advantage: No high voltage, no tritium, potentially very compact.

7. Conclusions

1. **Physics is honest:** Standard Gamow gives $Q \sim 10^{-57}$ at room temperature. No energy gain is possible.
2. **The anomaly is extraordinary:** A 40-order-of-magnitude gap requires either new physics, unrecognized enhancements, or systematic error.
3. **Hot spots are the most plausible explanation:** X-ray-induced local heating to $\sim 3,500\text{K}$ could explain the observations without invoking new physics.

4. **Independent replication is critical:** The NASA result must be reproduced with proper controls before investing in scale-up.
5. **Pivot to neutron source:** If the anomaly is real, LCF's value is as a compact neutron generator, not an energy source.
6. **(New, 2026-07-07) The recommended next step, by expected information gain per dollar, is the Hydrogen Control experiment** — cheaper and more discriminating than the NASA Replication or Temperature Mapping studies emphasized above. See §5.1.

Appendix A: Spark Engine Modules

Module	Description
physics	Gamow integration, Q factor, screening
lattice	DPA tracking, lifetime model
anomaly	Gap analysis framework
literature	11 canonical observations
hypothesis_models	Hot spot, phonon, screening models
rate_gap	Quantitative gap calculator
experimental_design	3-phase, 9-experiment program
uncertainty	Monte Carlo propagation
neutron_source	Compact source design
shielding	Dose calculations
bayesian <i>(added 2026-07-07)</i>	Probability distribution over the 5 hypotheses, Bayes updates from observations
optimal_experiment <i>(added 2026-07-07)</i>	Expected-information-gain ranking, budget-constrained experiment sequencing

Appendix B: Running the Analysis

```
# Full analysis report
cargo run --example analysis_runner

# Live discriminating-experiment ranking (added 2026-07-07)
cargo run --example optimal_experiment_proposal 500000

# CLI tool
cargo run --bin spark -- analyze --rate 1000 --material Er
cargo run --bin spark -- hypotheses
```



```
cargo run --bin spark -- experiments --phase 1
cargo run --bin spark -- gamow --temp 1000 --screening 309
```

Appendix C: Test Coverage

- 99 unit tests (up from 42 at original draft), 1 doc test — verified 2026-07-07
- All passing; CI (GitHub Actions) enforces `cargo fmt --check` and `cargo clippy -- -D warnings` on every push

Appendix D: References

- Steinetz, B. M. et al. (2020). "Novel nuclear reactions observed in bremsstrahlung-irradiated deuterated metals." *Physical Review C*, 101(4), 044610.
- Raiola, F. et al. (2004). "Enhanced electron screening in d(d,p)t for deuterated metals." *European Physical Journal A*, 19(2), 283–287.
- Czerski, K. et al. (2004). "Enhancement of the electron screening effect for d + d fusion reactions in metallic environments." *Europhysics Letters*, 68(3), 363.
- Huizenga, J. R. (1993). *Cold Fusion: The Scientific Fiasco of the Century*. University of Rochester Press. (Skeptical review cited in §4.2 as the field's documented failure modes.)
- Brown, D. A. et al. (2018). "ENDF/B-VIII.0: The 8th major release of the nuclear reaction data library." *Nuclear Data Sheets*, 148, 1–142. (Neutron cross-section validation source, per `literature_validation.rs`.)
- Miracle, D. B. & Senkov, O. N. (2017). "A critical review of high entropy alloys and related concepts." *Acta Materialia*, 122, 448–511. (HEA property validation source.)

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